AIORI HACKATHON 3

INNOVATION FOR A SECURE AND CONNECTED FUTURE

Problem Statement : PS-004

Authenticated Gateway-Assisted DNSSEC for IoT Clients



Team ID:
A3-PS004-TC133

Team Members:
Nethra Suresh
(nethrasuresh26@gmail.com)
N Yashaswi
(nyashaswi1505@gmail.com)

Team Name:
DNSentinal

ABSTRACT & PROBLEM DESCRIPTION

ABSTRACT

IoT devices rely on the Domain Name System (DNS) to locate online services, but conventional DNS does not independently guarantee the authenticity and integrity of DNS responses. DNS Security Extensions (DNSSEC) address this limitation through digital signatures and a chain of trust. However, performing complete DNSSEC validation can be challenging for resource-constrained IoT devices due to their limited processing power, memory, and energy.

This project proposes an **Authenticated Gateway-Assisted DNSSEC Validation System** in which a gateway performs DNSSEC validation on behalf of IoT clients and securely communicates the validation result

PROBLEM DESCRIPTION

- **Limited Resources:** IoT devices have limited memory and processing power.
- **Security Risks:** Fake DNS responses can redirect devices to malicious websites.
- **High Computation:** DNSSEC verification requires additional processing.
- **Gateway Trust:** IoT devices must verify that the gateway's response is authentic.

OBJECTIVE

To develop a gateway-assisted DNSSEC validation system that reduces the computational burden on resource-constrained IoT devices while ensuring secure and authenticated DNS responses.

BACKGROUND

DNS

The Domain Name System (DNS) translates domain names (e.g. www.google.com) into IP addresses, allowing devices to locate online services.

DNSSEC

DNS Security Extensions (DNSSEC) use digital signatures to provide authentication and integrity for DNS data, establishing a chain of trust from root to the domain

WHY IS IT IMPORTANT FOR IOT ?

- IoT Devices often have limited memory, processing power and energy
- Untrusted DNS responses may redirect devices to malicious destinations

OUR APPROACH

A trusted gateway performs DNSSEC validation on behalf of resource-constrained IoT clients and returns an authenticated validation result, reducing client-side computational burden while maintaining trust and security.

LITERATURE & RFC REVIEW

LITERATURE SURVEY

1. Beyond PKI: A DNSSEC Delegation Approach for Scalable Dynamic Credential Management in IoT
Explores DNSSEC-based delegation for scalable credential management and trust establishment in IoT devices.

2. Analysis and Evaluation of Post-Quantum Cryptography for DNSSEC (2025)
Evaluates post-quantum cryptographic algorithms for DNSSEC, including performance metrics such as latency and error rates.

3. Evaluating DNSSEC Against DNS Spoofing/MITM Attack in Virtualized Networks
Studies DNS spoofing and man-in-the-middle attacks and assesses DNSSEC-based protection in virtualized environments.

4. Proving DNSSEC Correctness: A Formal Approach to Secure Domain Name Resolution
Formally analyses DNSSEC validation, trust chains, and cache-related edge cases to examine protocol correctness.

RFC REVIEW

1.RFC 4033 — DNS Security Introduction and Requirements
Defines the security goals, concepts, and limitations of DNSSEC.

2.RFC 4034 — Resource Records for DNS Security Extensions
Specifies DNSSEC resource records such as DNSKEY, DS, RRSIG, and NSEC.

3.RFC 4035 — Protocol Modifications for DNSSEC
Defines the protocol changes required for DNSSEC validation and authenticated DNS responses.

4.RFC 7228 — Terminology for Constrained-Node Networks
Provides terminology and characteristics of constrained devices relevant to IoT clients.

PROPOSED SOLUTION

OVERVIEW

An authenticated gateway-assisted DNSSEC validation system for resource-constrained IoT devices, where a trusted gateway performs DNSSEC validation and returns an authenticated result to the client.

KEY FEATURES

- Gateway Validation : Performs full DNSSEC validation
- Authenticated Communication : Ensures integrity and authenticity of results
- Safe Failure Handling: Does not blindly trust uncertain results

Proposed Approach

- Gateway-Based Validation
  A trusted gateway performs DNSSEC validation on behalf of IoT clients
- Authenticated Results
  Validation results are sent through a secure and authenticated communication channel
- Lightweight Clients
  IoT devices avoid the computational burden of full DNSSEC validation
- Security-State Handling
  The system distinguishes secure, insecure, bogus and intermediate responses, and handles unverifiable results safely

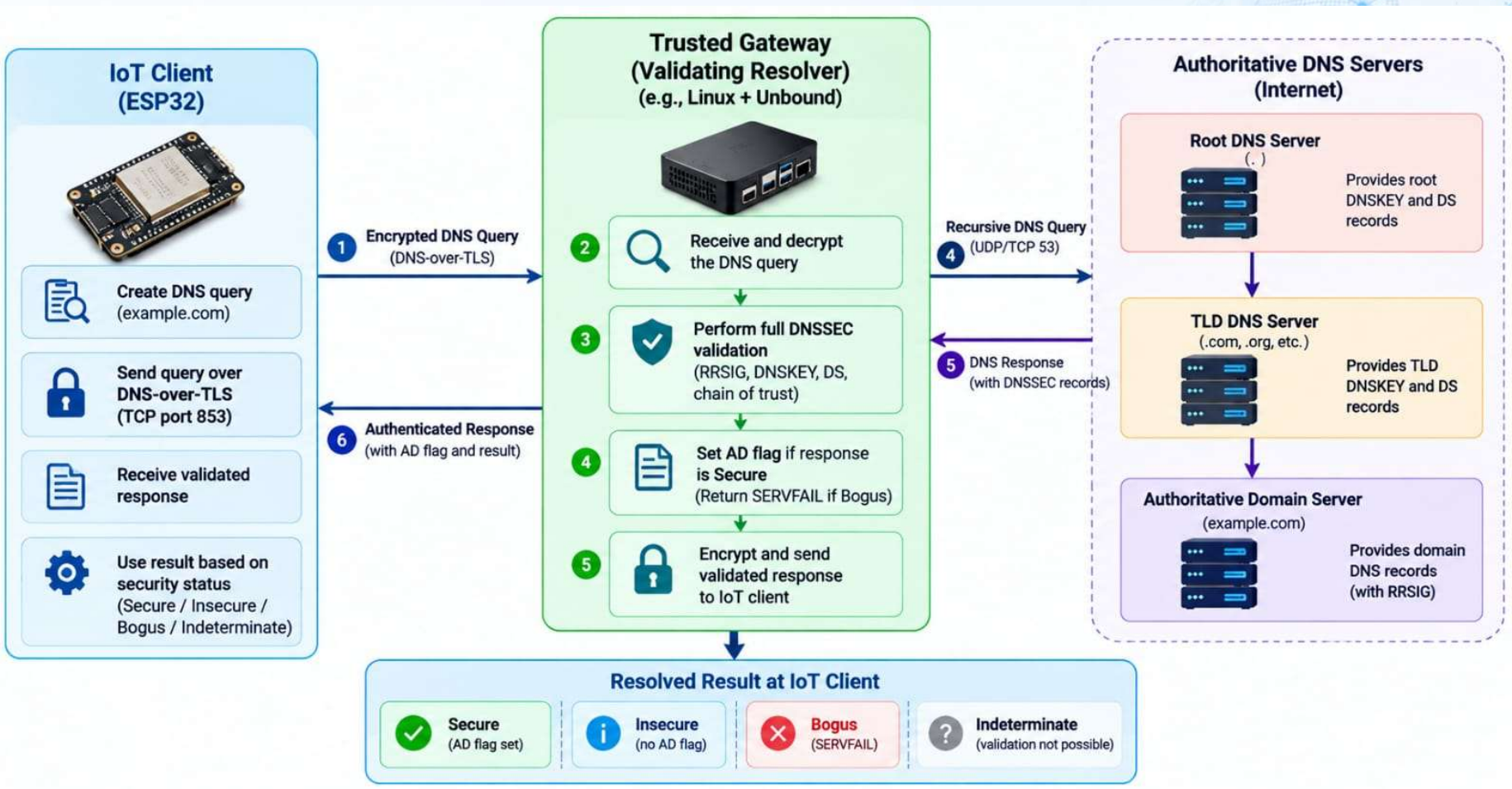
DNS-over-TLS (DoT)

Three Validation test cases

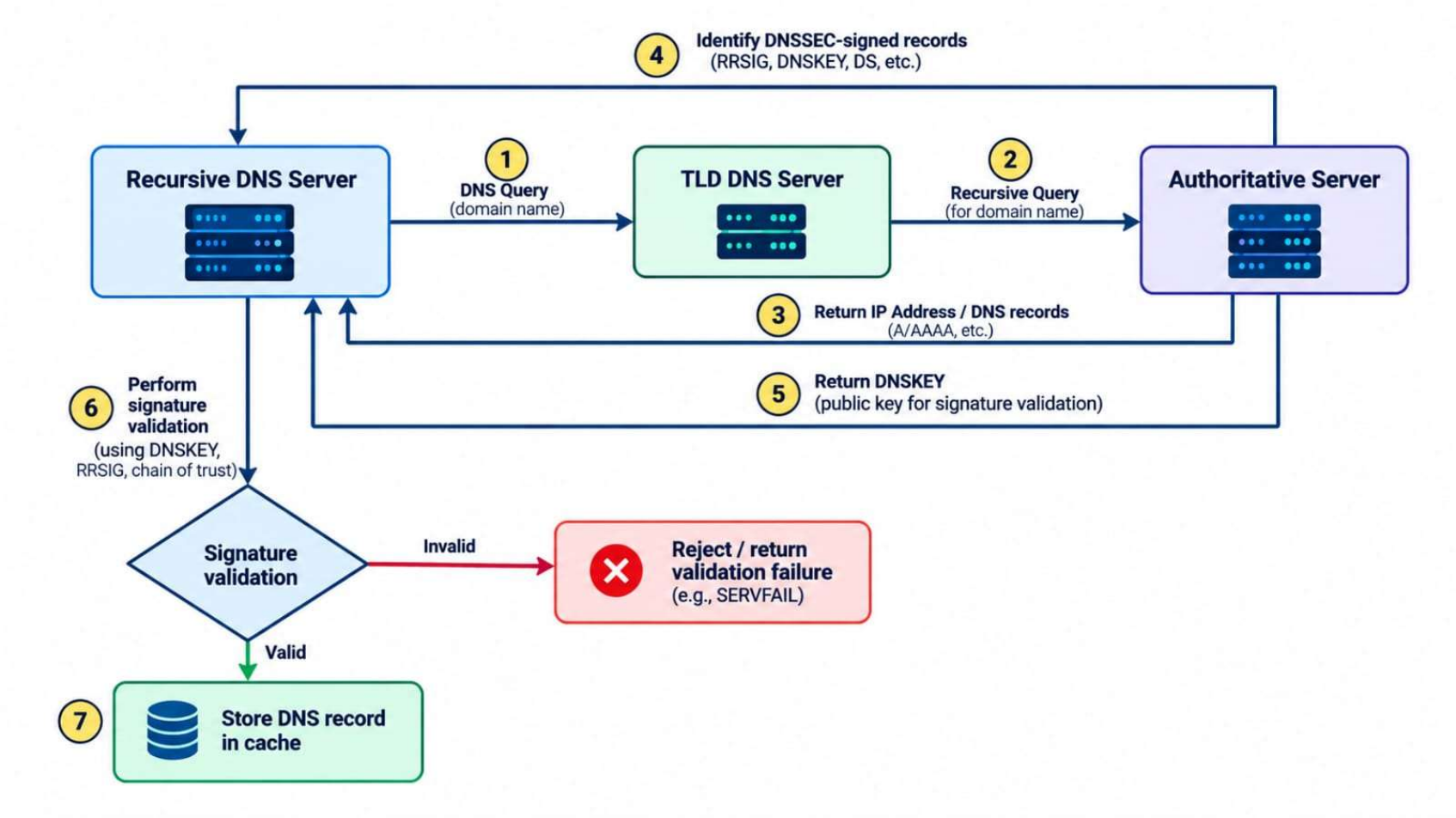
OPTIMIZATION PROPOSED

ASPECT	CONVENTIONAL APPROACH	PROPOSED APPROACH
1. DNSSEC Validation	Client performs full DNSSEC validation	Trusted Gateway performs full DNSSEC validation
2. Client Computation	High computational workload and cryptographic processing on IoT device	Reduced client-side workload , IoT device only sends query and processes authenticated result.
3. Memory Requirements	Large memory needed for parsing signatures, DNSSEC/DS records and trust anchors.	Minimal memory on client (only TLS buffer and basic DNS processing)
4. Communication Security	May use unencrypted DNS (vulnerable to eavesdropping and tampering)	Encrypted and authenticated communication using DNS-over TLS (DoT) on TCP port 853
5. Security Handling	Depends on client implementation and validation capability.	Gateway enforces validation and returns security status (Secure / Insecure / Bogus / Intermediate)
6. Time Synchronization	Requires accurate RTC or NTP on the IoT for signature time validation	Gateway handles NTP synchronization
7. Performance	High latency, CPU usage and energy consumption on IoT device	Lower client latency, reduced CPU cycles and energy usage due to offloaded validation

SOLUTION ARCHITECTURE & DESIGN



DATA FLOW DIAGRAM



DEPLOYMENT TIMELINE

PHASE 1

Infrastructure Deployment

- Setting up the development environment and gateway infrastructure

PHASE 2

Client Development

- Implement DNS query on ESP32
- Establish authenticated connection using DoT
- Implement response parsing and security status handling

PHASE 3

Integration and Security Testing

- Test with secure, insecure and bogus domain names
- Verify gateway authentication and certificate validation
- Debug and fix issues

PHASE 4

Performance Evaluation

- Measuring query latency, memory usage, CPU cycles.
- Compare with baseline approach (e.g. plain DNS / client-side DNSSEC)
- Final demonstration

BIBLIOGRAPHY

[1]D. Díaz-Sánchez *et al.*, “Beyond PKI: A DNSSEC Delegation Approach for Scalable Dynamic Credential Management in IoT,” *IEEE Internet of Things Journal*, vol. 12, no. 21, Nov. 2025.

[2] Nal and M. Niemiec, “Analysis and Evaluation of Post-Quantum Cryptography for DNSSEC,” *International Journal of Information and Communication Technology*, 2025.

[3] N. Vorn and T. K.-K. Jung, “Evaluating DNSSEC Against DNS Spoofing/MITM Attack in Virtualized Networks,” *International Journal of Content*, vol. 21, no. 3, Sep. 2025.

[4] Q. Zhang *et al.*, “Proving DNSSEC Correctness: A Formal Approach to Secure Domain Name Resolution,” arXiv preprint arXiv:2512.11431, Dec. 2025.

[5] Hoffman, P. (2023). *DNS Security Extensions (DNSSEC).* RFC 9364, IETF. https://www.rfc-editor.org/rfc/rfc9364

[6] NLnet Labs. *Unbound DNS Resolver Documentation*. https://unbound.docs.nlnetlabs.nl/

[7] Supporting constrained devices and networks in the decentralised Internet of Things. (2025). Internet of Things, Volume 30, Article 101496. https://doi.org/10.1016/j.iot.2025.101496

THANK YOU

